Module 01 : Introduction to recommendation systems

Q1. Explain recommendation system, its types, advantages.

A recommendation system is an intelligent software system that analyzes user data such as ratings, browsing history, purchase history, and preferences to recommend relevant items. It predicts the preferences or interests of users and suggests relevant items to them.

It answers questions like:

- Which movie should the user watch?
- Which product is likely to be purchased?
- Which song matches the user's taste?

Need for Recommendation System : -

With the rapid growth of digital content, users face difficulty in selecting relevant information from huge datasets.

Recommendation systems help by:

- Filtering useful content
- Personalizing user experience
- Saving search time
- Improving decision-making

Types of Recommendation Systems

Recommendation systems are broadly classified into the following types:

1. Collaborative Filtering Recommendation System :

Collaborative filtering recommends items based on the preferences of similar users.

It works on the assumption:

"Users who agreed in the past will agree in the future."

Working :

- Collect user ratings
- Find similar users/items
- Recommend items liked by similar users

Example :

If User A and User B have similar movie preferences, and User A liked a movie that User B has not watched, that movie is recommended to User B.

Types of Collaborative Filtering :

a) User-Based Collaborative Filtering - Finds users similar to the target user.

b) Item-Based Collaborative Filtering - Finds items similar to the items liked by the user.

Advantages

- No need for item content information
- Provides good personalized recommendations

Disadvantages

- Cold start problem
- Data sparsity

2. Content-Based Recommendation System :

Content-based filtering recommends items similar to those the user liked previously.

It uses item features such as:

- Genre
- Keywords
- Tags
- Description

Working

- Build item profile

- Build user profile
- Match both profiles

Example :

If a user watches action movies, the system recommends other action movies.

Advantages

- No dependency on other users
- Personalized suggestions

Disadvantages

- Limited diversity
- Overspecialization

3. Knowledge-Based Recommendation System :

Knowledge-based recommendation systems recommend items using domain knowledge and user requirements.

They use:

- Constraints
- Rules
- Explicit user preferences

Working :

The system asks users questions and recommends accordingly.

Example :

Laptop recommendation based on:

- Budget
- RAM
- Processor
- Usage purpose

Advantages

- Useful when historical data is unavailable
- Effective for expensive or infrequent purchases

Disadvantages

- Requires knowledge engineering
- Complex implementation

4. Hybrid Recommendation System :

Hybrid systems combine two or more recommendation techniques to improve performance.

Example :

Netflix combines collaborative and content-based filtering.

Advantages

- Better accuracy
- Reduces weaknesses of individual systems

Disadvantages

- Higher computational complexity
- Difficult to design

Advantages of Recommendation System :

1. Personalization - Provides customized suggestions based on user interests.
   Example: Personalized movie recommendations.
2. Reduces Information Overload - Helps users find relevant items quickly from large datasets.
3. Improves User Experience - Makes platforms more interactive and user-friendly.
4. Increases Sales and Revenue - Recommends products that users are likely to purchase.
   Example: "Customers who bought this also bought…"

5. Saves Time - Users do not need to search manually.
6. Enhances Customer Satisfaction - Relevant recommendations improve trust and engagement.
7. Supports Decision Making - Helps users choose suitable products or services.
8. Increases User Retention - Keeps users engaged on platforms.

Limitations of Recommendation System :

- Cold start problem
- Data sparsity
- Privacy concerns
- Scalability challenges
- Bias in recommendations

Applications of Recommendation Systems :

- **E-commerce** → Product recommendation
- **Entertainment** → Movies and music
- **Education** → Course recommendation
- **Social media** → Friend/content suggestion
- **News portals** → Article recommendation

Q2. What is meant by eliciting ratings in recommender systems?
Eliciting Ratings is the process of obtaining user preferences, opinions, or feedback about items in a recommender system.
The ratings collected help the system:

- Understand user interests
- Predict future preferences
- Recommend relevant items

Need for Eliciting Ratings
Recommendation systems require user data to function effectively. Without ratings or feedback, the system cannot identify:

- Which items users like
- Which items users dislike
- Similarity between users
- Similarity between items

Thus, eliciting ratings is essential for personalization and recommendation accuracy.

Types of Ratings in Recommender Systems
Ratings are mainly classified into two categories:

1. Explicit Ratings
Explicit ratings are directly provided by users intentionally. Users clearly express their opinion regarding an item.
Examples

- Rating a movie from 1 to 5 stars
- Like/Dislike buttons
- Product reviews
- Numerical scores

Example:

- A user gives 5 stars to a movie on Netflix.

Characteristics of Explicit Ratings

- Direct feedback
- Easy to interpret
- High accuracy
- Requires user effort

Advantages :
1. Clear User Preference - The system gets exact information about likes and dislikes.
2. Better Recommendation Accuracy - Explicit ratings improve prediction quality.
3. Easy to Analyze - Numerical values are easy to process mathematically.

Disadvantages :
1. Users May Not Provide Ratings - Many users ignore rating requests.
2. Sparse Data Problem - Only a small number of items receive ratings.
3. User Bias - Ratings may differ due to personal habits.

2. Implicit Ratings

Implicit ratings are collected indirectly from user behavior without asking users explicitly. The system observes user activities and derives preferences automatically.

Examples

- Purchase history
- Browsing history
- Watch time
- Clicks
- Search history
- Time spent on an item

Example:

If a user watches action movies repeatedly on YouTube, the system assumes the user likes action content.

Characteristics of Implicit Ratings

- Collected automatically
- No extra effort from users
- Large amount of data available
- Preference interpretation may be uncertain

Advantages :
1. Easy Data Collection - No need to ask users directly.
2. Large Dataset Availability - Continuous user activity generates more data.
3. Reduces User Burden - Users need not manually rate items.

Disadvantages
1. Ambiguous Preferences - Clicks do not always indicate liking.
2. Noise in Data - User actions may not represent actual interest.
3. Lower Accuracy - Inference may sometimes be incorrect.

Methods for Eliciting Ratings

Recommendation systems use different techniques to collect ratings:

1. Rating Scales - Users assign numerical values such as:

- 1 to 5 stars
- 1 to 10 ratings

Example:

Movie rating systems.

2. Like/Dislike Mechanism - Users simply approve or reject items.

Example: Thumbs up/down system.

3. Reviews and Comments - Users provide textual feedback about items.

4. Behavioral Tracking - The system monitors:

- Clicks
- Purchases

- Viewing history
- Search patterns

5. Pairwise Comparison - Users compare two items and choose the preferred one.

Challenges in Eliciting Ratings :

1. Cold Start Problem - New users may not provide enough ratings initially.
2. Data Sparsity - Most users rate only a few items.
3. Inconsistent Ratings - Different users interpret rating scales differently.
4. Privacy Concerns - Users may hesitate to share preferences.
5. Fake Ratings and Attacks - Some users may intentionally provide misleading ratings.

Importance of Eliciting Ratings

Eliciting ratings helps in:

- Personalization
- Improving recommendation quality
- Understanding user behavior
- Predicting future preferences
- Increasing customer satisfaction

Q3. What is a covariance matrix and why is it used? Explain how covariance matrices helps in understanding relationships between users or items.

Covariance is a statistical measure that indicates the direction of relationship between two variables.

- Positive covariance → variables increase together
- Negative covariance → one increases while the other decreases
- Zero covariance → no relationship

Mathematically:

$$Cov(X,Y) = \frac{\sum_{i=1}^{n}(X_i-\bar{X})(Y_i-\bar{Y})}{n-1}$$

Where:

- X and Y are variables
- $\bar{X}$ and $\bar{Y}$ are mean values
- n is number of observations

Covariance Matrix

A covariance matrix is a square matrix that contains covariance values between multiple variables.

It shows:

- Variance of variables on diagonal elements
- Covariance between variables on off-diagonal elements

General form:

$$C = \begin{bmatrix} Cov(X_1,X_1) & Cov(X_1,X_2) & \cdots \\ Cov(X_2,X_1) & Cov(X_2,X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Why Covariance Matrix is Used :

Covariance matrices are used because they help analyze relationships between variables in multidimensional data.

1. Identifying Similarity - Helps determine similarity between users or items.
2. Understanding Data Patterns - Shows how ratings vary together.
3. Dimensionality Reduction - Used in machine learning techniques like Principal Component Analysis (PCA).
4. Recommendation Generation - Improves collaborative filtering recommendations.
5. Feature Analysis - Helps identify related features in datasets.

Use of Covariance Matrix in Recommendation Systems :

In recommender systems, covariance matrices help understand relationships among:

- Users
- Items
- Ratings

1. User-User Relationship :

The covariance matrix measures similarity between users based on ratings.
Example:
If two users consistently rate movies similarly, covariance will be positive.
This helps in:

- User-based collaborative filtering
- Finding nearest neighbours

Example :

User A and User B both like action movies and rate them highly.
Positive covariance indicates similar taste.
Hence, movies liked by User A can be recommended to User B.

2. Item-Item Relationship :

Covariance matrices also identify relationships between items.
Example:
If users who buy laptops also buy laptop bags, covariance between these items becomes positive.
This helps in:

- Item-based recommendation systems
- Product association analysis

3. Detecting Preference Patterns :

Covariance helps discover hidden user interests and behavioural patterns.
Example:
Users preferring science fiction movies may also like fantasy movies.

4. Clustering Similar Users or Items :

Users/items with similar covariance values can be grouped into clusters.
Applications:

- Market segmentation
- Personalized recommendation

Advantages of Covariance Matrix :

1. Measures Relationships Efficiently - Provides mathematical representation of dependencies.
2. Useful in Collaborative Filtering - Helps identify similar users and items.
3. Supports Machine Learning Algorithms - Widely used in statistical and predictive models.
4. Helps in Pattern Recognition - Detects trends and behavioural similarities.
5. Simplifies Complex Data Analysis - Represents multidimensional relationships compactly.

Limitations of Covariance Matrix :

1. Scale Dependency - Covariance values depend on data scale.
2. Difficult Interpretation - Large covariance values may not always imply strong relationships.
3. Sensitive to Outliers - Extreme values can distort covariance.
4. Does Not Measure Strength Precisely - Correlation is often preferred for standardized comparison.

Applications of Covariance Matrix :

- Recommendation systems
- Financial risk analysis

- Machine learning
- Pattern recognition
- Data mining
- Image processing

~~Q4.~~ What are the different evaluation metrics for RS?
Evaluation metrics are used to measure the performance and effectiveness of a Recommendation System (RS). These metrics help determine how accurately and efficiently the system recommends relevant items to users. Evaluation is important because a good recommender system should not only provide accurate recommendations but also ensure user satisfaction, diversity, scalability, and robustness.

Evaluation metrics are used to:

- Measure recommendation accuracy
- Compare different recommendation algorithms
- Improve system performance
- Analyze user satisfaction
- Check scalability and robustness

Types of Evaluation Metrics in Recommendation Systems

1. Accuracy Metrics

Accuracy metrics measure how close predicted ratings are to actual user ratings.

These are the most commonly used evaluation metrics.

a) Mean Absolute Error (MAE)

MAE measures the average absolute difference between predicted ratings and actual ratings.

Formula:

$$MAE = \frac{1}{N} \sum_{i=1}^{N} |P_i - A_i|$$

Where:

- P_i = Predicted rating
- A_i = Actual rating
- N = Number of predictions

Lower MAE indicates better recommendation accuracy.

Advantages

- Simple to calculate
- Easy to understand

b) Root Mean Square Error (RMSE)

RMSE measures square root of average squared differences between predicted and actual ratings.

Formula:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (P_i - A_i)^2}$$

Advantages

- Penalizes large errors more heavily
- Provides better accuracy measurement

Disadvantages

- Sensitive to outliers

c) Mean Squared Error (MSE)

MSE calculates average squared prediction error.

Formula:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(P_i - A_i)^2$$

Lower MSE means better prediction quality.

2. Decision-Support Metrics

These metrics evaluate how effectively the system helps users make decisions.

a) Precision

Precision measures the proportion of recommended items that are relevant.

Formula:

$$Precision = \frac{Relevant\ Items\ Recommended}{Total\ Recommended\ Items}$$

Example
If 8 out of 10 recommended movies are relevant:

Precision = 8/10 = 0.8

Advantages

- Measures recommendation quality

Disadvantages

- Ignores unrecommended relevant items

b) Recall

Recall measures the proportion of relevant items successfully recommended.

Formula:

$$Recall = \frac{Relevant\ Items\ Recommended}{Total\ Relevant\ Items}$$

Advantages

- Measures coverage of relevant items

Disadvantages

- High recall may reduce precision

c) F1-Score

F1-score combines precision and recall.

Formula:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Advantages

- Balanced evaluation metric

Disadvantages

- Difficult interpretation in some cases

3. Coverage Metrics

Coverage measures the percentage of items or users for which recommendations can be generated.

Types

- Item coverage
- User coverage

Advantages

- Indicates system usefulness

Disadvantages

- High coverage may reduce accuracy

4. Confidence and Trust Metrics

These metrics evaluate how much users trust the recommendations.

Factors affecting trust:

- Recommendation transparency
- Recommendation consistency
- Explanation quality

5. Novelty Metrics

Novelty measures whether recommendations are new or unknown to users.

Example

Suggesting less popular but useful movies.

Advantages

- Enhances discovery experience

Disadvantages

- Very novel items may not always be relevant

6. Serendipity Metrics

Serendipity measures how surprising and useful recommendations are.

Example

A user unexpectedly likes a recommended song from a different genre.

Advantages

- Increases user satisfaction
- Encourages exploration

7. Diversity Metrics

Diversity measures how varied the recommendations are.

Example

Recommending movies from different genres instead of only action movies.

Advantages

- Prevents repetitive recommendations
- Improves user experience

Disadvantages

- Too much diversity may reduce relevance

8. Robustness Metrics

Robustness measures the ability of the recommendation system to resist attacks or fake ratings.

Example

Protection against spam reviews or manipulated ratings.

Q5. Explain the concept of inverse matrix and its importance.

An inverse matrix is a matrix that when multiplied with the original matrix gives an identity matrix.

If A is a square matrix, then its inverse is represented by A^{-1}.

Condition:

$$A \times A^{-1} = A^{-1} \times A = I$$

Where:

- A = Original matrix
- A^{-1} = Inverse matrix
- I = Identity matrix

Identity Matrix

An identity matrix is a square matrix in which:

- Diagonal elements are 1
- All other elements are 0

Example:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Conditions for Existence of Inverse Matrix

A matrix has an inverse only if:

1. Matrix Must Be Square

Number of rows must equal number of columns.

2. Determinant Must Not Be Zero

Condition:

$$|A| \neq 0$$

If determinant is zero, the matrix is called a singular matrix and inverse does not exist.

Formula for Inverse of a 2×2 Matrix

For matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Inverse is:

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Where:

$$ad - bc \neq 0$$

Importance of Inverse Matrix :

1. Solving Linear Equations

Inverse matrices are used to solve systems of linear equations.

Equation:

AX=B

Solution:

$X=A^{-1}B$

This method is widely used in engineering and scientific computations.

2. Used in Recommendation Systems

In recommendation systems, matrix operations are used for:

- User-item rating matrices
- Collaborative filtering
- Matrix factorization

Inverse matrices help in optimization and data transformations.

3. Computer Graphics and Image Processing
Used for:

- Rotation
- Scaling
- Transformations
- Image reconstruction

4. Machine Learning and Data Science
Inverse matrices are used in:

- Linear regression
- Covariance matrix computations
- Statistical modeling

5. Cryptography - Used in encryption and decryption algorithms.
6. Network Analysis - Helps in solving flow and connectivity problems.

Q6. What is the role of user – item matrix interaction matrix in RS.

A User–Item Interaction Matrix is a matrix representation where:

- Rows represent users
- Columns represent items
- Matrix values represent interactions or ratings

General representation:

Users / Items	Item 1	Item 2	Item 3
User 1	5	3	?
User 2	4	?	2
User 3	?	5	4

Where:

- Numerical values represent ratings/interactions
- "?" indicates missing or unknown interactions

Types of User–Item Interactions

1. Explicit Interaction

Users directly provide ratings or feedback.

Examples:

- 5-star movie rating
- Like/Dislike button
- Product review

2. Implicit Interaction

User preferences are collected indirectly through behaviour.

Examples:

- Purchase history
- Watch time
- Clicks
- Search history

Role of User–Item Interaction Matrix :

1. Stores User Preferences

The matrix records user interactions with items such as ratings, purchases, and views. This helps the system understand user interests and behaviour.

2. Helps in Collaborative Filtering

It is the basic input for collaborative filtering algorithms. The system uses the matrix to identify similar users or similar items for generating recommendations.

3. Predicts Missing Ratings

Many entries in the matrix are missing because users rate only a few items. Recommendation systems predict these missing values to suggest relevant items.

4. Supports Personalized Recommendations

The matrix enables the system to recommend items according to individual user preferences and past interactions.

5. Helps in Similarity Computation

Using the interaction data, the system calculates similarity between:

- User-to-user
- Item-to-item

This improves recommendation accuracy.

Q7. How does matrix factorization contribute to RS.

Matrix Factorization is a mathematical technique that decomposes a large user–item matrix into two lower-dimensional matrices.

If R is the user–item rating matrix, then:

$$R \approx P \times Q^T$$

Where:

- R = User–Item Rating Matrix
- P = User Feature Matrix
- Q = Item Feature Matrix
- Q^T = Transpose of item matrix

The smaller matrices represent hidden or latent features of users and items.

Working of Matrix Factorization

Step 1: Create User–Item Matrix

Example:

Users / Movies	Movie A	Movie B	Movie C
User 1	5	?	3
User 2	4	2	?
User 3	?	5	4

The matrix contains many missing values.

Step 2: Factorize the Matrix

The matrix is decomposed into:

- User latent feature matrix
- Item latent feature matrix

Step 3: Predict Missing Values

The product of the smaller matrices predicts missing ratings.

Example:
If a user likes action movies and an item belongs to the action genre, the predicted rating becomes high.

Contribution of Matrix Factorization in Recommendation Systems

1. Predicts Missing Ratings

Matrix factorization estimates unknown ratings in sparse matrices.

This helps recommend items that users are likely to prefer.

2. Handles Data Sparsity

In real-world systems, users rate only a few items.

Matrix factorization reduces sparsity by discovering hidden relationships between users and items.

3. Identifies Latent Features

It discovers hidden factors such as:

- Movie genre preference
- Music taste
- Product category interest

These hidden features improve recommendation quality.

4. Improves Recommendation Accuracy

By learning hidden patterns from data, matrix factorization generates highly accurate personalized recommendations.

5. Supports Scalability

Matrix factorization works efficiently with large datasets containing millions of users and items.

Hence, it is suitable for large-scale systems like:

- Netflix ↗
- Amazon ↗
- Spotify ↗

Types of Matrix Factorization Techniques

1. Singular Value Decomposition (SVD)

SVD decomposes the matrix into three matrices and is widely used in collaborative filtering.

2. Non-negative Matrix Factorization (NMF)

All matrix values remain non-negative, making interpretation easier.

3. Probabilistic Matrix Factorization (PMF)

Uses probability distributions for recommendation prediction.

Q8. Compare Implicit and Explicit RS.

Basis	Explicit RS	Implicit RS
Definition	Collects direct feedback from users	Collects indirect feedback from user behaviour
User Input	Ratings, reviews, likes	Clicks, views, purchases, watch time
Data Collection	Manual	Automatic
Accuracy of Preference	Usually high	May be uncertain
User Effort	Requires user participation	No extra user effort
Data Availability	Limited data	Large amount of data
Interpretation	Easy to interpret	Difficult to interpret
Examples	Star ratings, reviews	Browsing history, search history
Sparsity Problem	More common	Less common
Reliability	More reliable	May contain noisy data

Q9. Why are unnary rating significantly different from other types of ratings in the design of RS.

Definition of Unary Ratings

Unary ratings are ratings in which only positive interactions are recorded.

The system knows:

- What the user interacted with
- But does not know what the user disliked

In unary ratings:

- Interaction is represented by 1
- No interaction is represented by 0 or missing value

Example:

- Purchase history
- Video watched
- Song played
- Product clicked

Example of Unary Rating Matrix

Users / Items	Movie A	Movie B	Movie C
User 1	1	0	1
User 2	1	1	0
User 3	0	1	1

Where:

- 1 = User interacted with item
- 0 = No interaction recorded

Why Unary Ratings Are Significantly Different

1. Lack of Negative Feedback

Unary ratings only provide positive interaction information.

The system does not know:

- Whether the user disliked the item
- Or simply never saw the item

This makes recommendation design more difficult.

Example

If a user did not watch a movie, the system cannot determine whether:

- The user dislikes it
- Or is unaware of it

2. Presence-Only Data

Unary systems only record whether interaction occurred.

Traditional rating systems provide:

- Degree of liking
- Preference intensity

Unary ratings do not provide rating strength.

3. Data Sparsity Challenges

Most items remain unobserved in unary datasets.

This creates highly sparse matrices which affect recommendation accuracy.

4. Ambiguity in Missing Values

In explicit ratings:

- Missing value often means "not rated."

In unary ratings:

- Missing value may mean:
  - Not interested
  - Not discovered
  - No opportunity to interact

Thus interpretation becomes difficult.

5. Different Learning Algorithms Required

Unary rating systems require specialized algorithms such as:

- Implicit collaborative filtering
- One-class collaborative filtering
- Probabilistic models

Traditional algorithms designed for explicit ratings may not work effectively.

Q10. Analyze the key challenges involved in developing effective recommender systems.

Recommendation Systems (RS) are used to suggest relevant items such as movies, products, music, and videos to users. Developing an effective recommender system is difficult because of issues related to data, scalability, privacy, and changing user behaviour.

Key Challenges in Recommendation Systems

1. Cold Start Problem

This occurs when new users or new items have very little data available. The system cannot generate accurate recommendations due to insufficient ratings or interactions.

2. Data Sparsity

Users usually interact with only a few items, leaving many missing values in the user–item matrix. This reduces the efficiency of collaborative filtering methods.

3. Scalability

Modern systems contain millions of users and items. Processing such large datasets requires high computational power and memory.

4. Accuracy vs Diversity Trade-off

Highly accurate recommendations may become repetitive. Lack of diversity reduces user interest and limits content exploration.

5. Overspecialization

Content-based systems often recommend only similar items. Users may miss new or different types of content.

6. Privacy and Security Issues

Recommendation systems collect personal user data such as browsing and purchase history. Protecting this information from misuse is a major challenge.

7. Shilling and Attack Problems

Fake users may provide manipulated ratings to increase or decrease item popularity. This affects recommendation quality and trustworthiness.

8. Dynamic User Preferences

User interests change over time. Recommendation systems must continuously update recommendations according to changing preferences.

9. Gray Sheep Problem

Some users have unique preferences that do not match any user group. It becomes difficult to find similar users for recommendations.

10. Synonym and Semantic Problems

Different items may have similar meanings but different names. This creates difficulty in identifying correct item similarity.

11. Evaluation Challenges

Measuring recommendation quality is difficult because factors like diversity, novelty, and user satisfaction must also be considered along with accuracy.

12. Real-Time Recommendation Challenges

Many applications require instant recommendations. Generating fast recommendations for large datasets is computationally challenging.

Q11. Discuss the important functions of a recommender system and their significance.
A Recommendation System (RS) suggests relevant items to users based on their preferences, ratings, and behaviour. It helps users find useful content from large datasets and improves personalization and user experience.

Important Functions of a Recommender System :

1. Personalized Recommendation
The system provides recommendations according to individual user interests and past behaviour.
Significance :
Improves user satisfaction and increases engagement.

2. Information Filtering
It filters useful information from a large amount of available data and removes irrelevant items.
Significance :
Reduces information overload and saves time.

3. Predicting User Preferences
The system predicts unknown user ratings or interests using past interactions and similarity analysis.
Significance :
Improves recommendation accuracy and future predictions.

4. Item Ranking
Items are ranked according to relevance and predicted usefulness for users.
Significance :
Displays the most useful recommendations first.

5. Similarity Identification
The system identifies similarities between users or items based on ratings and behaviour.
Significance :
Supports collaborative filtering and recommendation generation.

6. Content Discovery
Recommendation systems help users discover new and interesting items.
Significance :
Improves novelty and encourages exploration.

7. User Profiling
The system creates user profiles using browsing history, ratings, and preferences.
Significance :
Enables better personalization and understanding of user behaviour.

8. Decision Support
It assists users in selecting suitable products, services, or content.
Significance :
Simplifies decision-making and improves confidence.

9. Increasing Business Revenue
Relevant recommendations encourage users to purchase or consume more content.
Significance :
Increases sales, profits, and customer retention.

10. Continuous Learning
Modern systems continuously update recommendations using new user interactions.
Significance :
Keeps recommendations relevant and updated.

Module 02 : Collaborative Filtering

~~Q~~1. How do recommendation systems handle the cold-start problem?

The Cold-Start Problem is one of the major challenges in Recommendation Systems (RS). It occurs when the system has insufficient data about new users or new items, making it difficult to generate accurate recommendations.

Since recommendation systems depend on user interactions, ratings, and history, lack of initial information reduces recommendation quality.

Types of Cold-Start Problems

1. New User Cold Start

Occurs when a new user joins the system and has not provided enough ratings or interactions.

2. New Item Cold Start

Occurs when a new item is added and has very few or no ratings.

3. System Cold Start

Occurs when the recommendation system is newly launched and has very little overall data.

Methods to Handle Cold-Start Problem

1. Collect Initial User Preferences

The system asks new users to rate some items or select interests during registration.

Benefit
Provides initial data for generating personalized recommendations.

2. Use Content-Based Filtering

Recommendations are generated using item features such as genre, tags, keywords, or descriptions instead of user ratings.

Benefit
Useful for recommending new items with no interaction history.

3. Hybrid Recommendation Systems

Hybrid systems combine collaborative and content-based filtering methods.

Benefit
Reduces dependence on only user ratings and improves recommendation accuracy.

4. Use Demographic Information

The system uses user information such as age, gender, location, or occupation to generate initial recommendations.

Benefit
Helps recommend items to new users with limited interaction data.

5. Popularity-Based Recommendation

The system recommends trending or highly popular items to new users.

Benefit
Provides reasonable recommendations even without personalized data.

6. Social Media and External Data

The system uses data from social networks, browsing history, or external platforms.

Benefit
Improves understanding of user interests quickly.

7. Implicit Feedback Collection

The system tracks user behaviour such as clicks, watch history, and purchases.
Benefit
Collects preference data automatically without requiring explicit ratings.

Advantages of Solving Cold-Start Problem

- Improves recommendation accuracy
- Enhances user experience
- Increases user engagement
- Helps new items gain visibility
- Improves customer retention

Limitations

- Initial recommendations may still be less accurate
- Collecting user information may raise privacy concerns
- Hybrid systems increase system complexity

Applications

Cold-start handling techniques are used in:

- Netflix ↗
- Amazon ↗
- Spotify ↗

Q2. Explain different attacks on collaborative filtering with an example.

Collaborative Filtering (CF) is a widely used recommendation technique that generates recommendations based on user ratings and behaviour. Since collaborative filtering depends heavily on user-provided data, it is vulnerable to various attacks where fake users intentionally manipulate ratings to influence recommendations.

These attacks are also called:

- Shilling attacks
- Profile injection attacks

The main goal of such attacks is to unfairly promote or demote items in the recommendation system.

Need for Attacks in Collaborative Filtering

Attackers may try to:

- Increase popularity of an item
- Reduce ranking of competitor items
- Manipulate recommendation results
- Influence customer decisions

Types of Attacks on Collaborative Filtering

1. Push Attack

In a push attack, fake users provide very high ratings to a target item to increase its recommendation score.
Example *:* An online seller creates fake accounts and gives 5-star ratings to their product on Amazon to make it appear more popular.
Impact

- Artificially increases item popularity
- Produces biased recommendations

2. Nuke Attack

In a nuke attack, attackers give very low ratings to a target item to decrease its ranking.
Example : Competitors give 1-star ratings to a newly released movie to reduce its recommendations on Netflix.
Impact

- Reduces visibility of the item
- Affects recommendation accuracy

3. Random Attack

Attackers assign random ratings to many items along with manipulated ratings for the target item.
Example : Fake users rate several products randomly and give extremely high ratings to one specific product.
Impact

- Difficult to detect
- Introduces noisy data

4. Average Attack

In this attack, fake profiles rate items close to their average ratings and highly rate the target item.
Example : Attackers study average movie ratings and create realistic-looking profiles to promote one movie.
Impact

- More effective and harder to identify
- Mimics normal user behaviour

5. Bandwagon Attack

Attackers rate popular items highly along with the target item to make fake profiles appear genuine.

Example : Fake users highly rate trending movies and also rate one target movie positively.
Impact

- Increases trustworthiness of fake profiles
- Manipulates collaborative filtering results

6. Segment Attack
Attackers target a specific group of users with similar preferences.
Example : A gaming product is promoted only among users interested in gaming products.
Impact

- Influences recommendations for a particular user segment

Example of Collaborative Filtering Attack

Suppose a movie recommendation system contains the following ratings:

User	Movie A	Movie B	Target Movie
User 1	5	4	2
User 2	4	5	1
Fake User	5	5	5

The fake user intentionally gives a high rating to the target movie to increase its recommendation score.

As collaborative filtering identifies similar users, the target movie may get recommended unfairly.

Q3. Evaluate the impact of sparsity in RS and methods to handle it.
Sparsity is one of the major challenges in Recommendation Systems (RS). It occurs when users interact with only a small number of items, leaving most entries in the User–Item Interaction Matrix empty.
Since recommendation systems depend on user ratings and interactions, sparse data reduces the system's ability to generate accurate recommendations.

Sparsity in Recommendation Systems

A sparse matrix contains many missing or unknown values.

Example:

Users / Items	Movie A	Movie B	Movie C
User 1	5	?	?
User 2	?	4	?
User 3	?	?	5

Most values are missing, making the matrix sparse.

Impact of Sparsity in Recommendation Systems
1. Reduced Recommendation Accuracy
With limited ratings, the system cannot properly identify user preferences or item similarities.

2. Difficulty in Finding Similar Users or Items
Collaborative filtering requires common ratings between users/items. Sparse data reduces similarity calculations.

3. Cold-Start Problem
New users and new items increase sparsity because very little interaction data is available initially.

4. Poor Personalization
Insufficient user data leads to less personalized recommendations.

5. Increased Computational Complexity
Sparse datasets require advanced techniques and more processing for prediction.

Methods to Handle Sparsity :
1. Matrix Factorization
Techniques like SVD decompose the sparse user–item matrix into smaller matrices and predict missing ratings. This improves recommendation accuracy.

2. Hybrid Recommendation Systems
Hybrid systems combine collaborative and content-based filtering methods. This reduces dependence on limited rating data.

3. Clustering Techniques
Users or items with similar behaviour are grouped together. This helps improve similarity calculation and recommendations.

4. Use of Implicit Feedback
The system uses clicks, purchases, watch history, and browsing activity as additional data. This increases available interaction information.

5. Dimensionality Reduction
Large sparse matrices are reduced into smaller representations. This improves computational efficiency and prediction performance.

6. Popularity-Based Recommendation
Popular or trending items are recommended when sufficient user data is unavailable. This helps generate basic recommendations for new users.

7. Association Rule Mining
The system identifies items frequently used or purchased together. This helps recommend related items even with sparse ratings.

Q4. Explain Collaborative Filtering in detail.
Collaborative Filtering is a recommendation technique that predicts user interests by collecting preferences and ratings from many users and identifying similarities among them.
It recommends:

- Items liked by similar users
  or
- Items similar to those already liked by the user

Working of Collaborative Filtering :
Step 1: Collect User Ratings
The system collects user interactions such as:

- Ratings
- Purchases
- Clicks
- Watch history

Step 2: Create User–Item Matrix

A matrix is formed where:

- Rows represent users
- Columns represent items
- Values represent ratings/interactions

Step 3: Find Similarities

The system calculates similarity between:

- Users
  or
- Items

Using methods such as:

- Cosine similarity
- Pearson correlation
- Covariance

Step 4: Generate Recommendations

Items liked by similar users are recommended to the target user.

Types of Collaborative Filtering

1. User-Based Collaborative Filtering

In this method, users with similar preferences are identified.

Working

- Find users similar to the target user
- Recommend items liked by similar users

Example

User	Movie A	Movie B	Movie C
User 1	5	4	?
User 2	5	5	4

Since both users like Movie A and B, Movie C may be recommended to User 1.

Advantages

- Simple to implement
- Personalized recommendations

Disadvantages

- Scalability issues for large datasets

2. Item-Based Collaborative Filtering

This method identifies items similar to items already liked by the user.

Working

- Find similar items
- Recommend related items

Example

If users who watch action movies also watch thriller movies, thriller movies are recommended.

Advantages

- Better scalability
- Stable recommendations

Disadvantages

- Requires sufficient item interaction data

Model-Based Collaborative Filtering

This approach uses machine learning and statistical models to predict user preferences.

Techniques

- Matrix factorization
- Clustering
- Neural networks

Advantages

- High accuracy
- Efficient for large datasets

Disadvantages

- Computational complexity

Advantages of Collaborative Filtering

1. No Need for Item Features – Recommendations are based on user interactions instead of item descriptions.
2. Personalized Recommendations – Provides user-specific suggestions based on preferences.
3. Discovers Hidden Interests – Can recommend unexpected but relevant items.
4. Domain Independent – Works for movies, music, products, books, and more.

Disadvantages of Collaborative Filtering

1. Cold Start Problem – New users and items have insufficient data for recommendations.
2. Data Sparsity – Most users rate only a few items, creating sparse matrices.
3. Scalability Issues – Large datasets increase computation time.
4. Vulnerable to Attacks – Fake ratings can manipulate recommendations.

~~Q5~~. Differentiate between user-based and item-based collaborative filtering

Basis	User-Based Collaborative Filtering	Item-Based Collaborative Filtering
Definition	Recommends items based on similar users	Recommends items based on similar items
Main Focus	Similarity between users	Similarity between items
Working Principle	Users with similar interests receive similar recommendations	Items similar to previously liked items are recommended
Similarity Calculation	User-to-user similarity	Item-to-item similarity
Data Used	User ratings and behaviour	Item ratings and relationships
Recommendation Process	Finds nearest neighbour users first	Finds similar items first
Example	If User A and User B like similar movies, movies liked by B are recommended to A	If a user likes an action movie, similar action movies are recommended
Scalability	Less scalable for large datasets	More scalable and efficient
Stability	User preferences may change frequently	Item similarities are more stable
Accuracy	Good for small systems	Better for large systems
Computational Cost	Higher due to changing user data	Lower because item similarities change less frequently
Sparsity Handling	Affected more by sparse data	Handles sparsity better
Real-World Usage	Less commonly used in large platforms	Widely used in modern recommendation systems

Q6. Discuss model-based collaborative filtering approaches, including their benefits over memory-based methods.

Model-based collaborative filtering builds a predictive model using user ratings and interaction data. The model learns hidden patterns and relationships between users and items to generate recommendations.

Instead of directly using the entire dataset during recommendation, the system first trains a model and then predicts ratings.

Working of Model-Based Collaborative Filtering

Step 1: Collect User–Item Data

The system gathers:

- Ratings
- Purchases
- Clicks
- Watch history

Step 2: Train the Model

Machine learning or statistical algorithms are used to identify hidden relationships in the data.

Step 3: Predict Missing Ratings

The trained model predicts unknown user preferences and recommends relevant items.

Approaches of Model-Based Collaborative Filtering :

1. Matrix Factorization

The user–item matrix is decomposed into smaller latent feature matrices.
Example : Singular Value Decomposition (SVD)
Benefit : Improves recommendation accuracy and handles sparse data effectively.

2. Clustering Methods

Users or items with similar behaviour are grouped into clusters.
Benefit : Reduces computation time and improves scalability.

3. Bayesian Models

Probability-based models are used to predict user preferences.
Benefit :Handles uncertainty and incomplete data efficiently.

4. Neural Network-Based Models

Deep learning techniques are used to learn complex user-item relationships.
Benefit : Provides highly personalized recommendations.

5. Association Rule Mining

Frequently associated items are identified using rule-based methods.
Benefit : Useful in market basket and product recommendation systems.

Benefits of Model-Based Methods Over Memory-Based Methods :

1. Better Scalability

Model-based methods work efficiently for large datasets with millions of users and items.

2. Higher Accuracy

Machine learning models identify hidden patterns and improve prediction quality.

3. Handles Sparsity Better

Matrix factorization and clustering reduce problems caused by missing ratings.

4. Faster Recommendation Generation
Once the model is trained, recommendations can be generated quickly.

5. Reduced Memory Usage
The entire dataset need not be processed repeatedly during recommendation.

6. Better Handling of Noisy Data
Statistical models reduce the effect of inconsistent or fake ratings.

~~Q7~~. Explain bandwagon attack with an example.
A Bandwagon Attack is an attack in which fake users give high ratings to:

- Popular items already liked by many users
- The target item that the attacker wants to promote

By associating the target item with popular items, the fake profiles appear genuine to the recommendation system.

Working of Bandwagon Attack :
Step 1: Identify Popular Items
Attackers first identify items that are already highly rated by many users.
Step 2: Create Fake Profiles
Fake user accounts are created with ratings similar to normal users.
Step 3: Rate the Target Item Highly
The fake users give very high ratings to the target item.
Step 4: Manipulate Recommendations
Collaborative filtering algorithms identify the fake profiles as similar users and begin recommending the target item to others.

Example of Bandwagon Attack

Suppose a movie recommendation system contains these ratings:

User	Popular Movie A	Popular Movie B	Target Movie
User 1	5	5	2
User 2	4	5	1
Fake User	5	5	5

The fake user gives:

- High ratings to popular movies
- A very high rating to the target movie

Since the fake profile matches common user behaviour, the recommendation system may start recommending the target movie to many users.

Characteristics of Bandwagon Attack :

- Uses popular items to appear realistic
- Difficult to detect
- Influences collaborative filtering systems
- Increases target item visibility

Impact of Bandwagon Attack :
1. Manipulates Recommendation Results
Target items receive unfairly high recommendations.
2. Reduces Recommendation Accuracy
The system generates biased recommendations.

3. Decreases User Trust
Users may lose confidence in the recommendation system.
4. Introduces Fake Data
Fake profiles increase noisy and misleading data.